VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"Jnana Sangama", Machhe, Belagavi, Karnataka-590018

A Project Synopsis on

"Online Quiz Platform with Performance Analysis and Leaderboards"

Submitted in partial fulfillment towards Project Work of VI Semester of

Bachelor of Engineering

in

Computer Science and Engineering

Submitted by

Hamsa R Gowda [4GW23CS035]
Manasa N [4GW23CS056]
Monika MU [4GW23CS062]
Megha HR [4GW24CS406]

Under the Guidance of

Mrs. Ashwini MS

Assistant Professor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(Accredited by NBA, New Delhi, Validity: 01.07.2023 – 30.06.2026)

GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN

(Affiliated to VTU, Belagavi, Approved by AICTE, New Delhi & Govt. of Karnataka)

(Accredited with Grade 'A' by NAAC)

K.R.S Road, Metagalli, Mysuru-570016, Karnataka

2025-2026

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Project Synopsis

Team Members	1. Hamsa R Gowda [4GW23CS035] 2. Manasa N [4GW23CS056] 3. Monika MU [4GW23CS062] 4. Megha HR [4GW24CS406]
Name of Guide	Mrs. Ashwini MS, Assistant Professor, Dept. of CSE
In House / Industry Project	In-House Project
Batch No.	Batch 22

Sl. No.	Particulars	Description
1.	**Title**	Online Quiz Platform with Performance Analysis and Leaderboards
2.	**Aim**	To develop an AI-powered adaptive quiz platform for personalized learning, intelligent assessment, and performance analytics.
3.	**Introduction**	The AI-Powered Adaptive Quiz and Performance Analytics Platform is an intelligent online assessment system designed to enhance digital learning through adaptive quizzes, real-time performance analytics, and personalized educational support. Unlike traditional quiz platforms, the system uses Artificial Intelligence, learning analytics, and gamification techniques to improve student engagement, identify strengths and weaknesses, and provide customized learning

		making it highly useful for schools, colleges, coaching institutes, and training organizations in modern digital education environments.
4.	**Problem Statement**	Traditional online quiz and assessment systems mainly focus on conducting tests and displaying scores, but they lack personalized learning, adaptive assessments, intelligent analytics, and secure evaluation mechanisms. Existing platforms provide limited support for tracking student performance, identifying strengths and weaknesses, offering AI-driven recommendations. Additionally, manual evaluation processes, limited engagement features, and inadequate anti-cheating mechanisms reduce the effectiveness and reliability of digital learning and assessment systems.
5.	**Objectives**	• To implement AI-powered adaptive quizzes that personalize learning based on student performance and behavior. • To provide detailed performance analytics and personalized recommendations for improving learning outcomes. • To ensure secure and engaging online assessments using anti-cheating mechanisms, real-time leaderboards, and gamification features. • To provide AI-driven predictive analytics that monitor student progress and generate smart revision plans for continuous academic improvement.
6.	**Software and Hardware Requirements**	**Software:** • OS: Windows 10/11 | IDE: VS Code | Version Control: Git & GitHub • Frontend: HTML, CSS, JavaScript, TypeScript, React.js, Vite • Backend: Python (for AI/ML modules), Supabase backend services, Node.js • Database: MongoDB, PostgreSQL

		Hardware: • CPU: Intel i3 / AMD Ryzen 5 or higher • RAM: Minimum 8 GB | Storage: 256 GB SSD minimum (512 GB preferred for dataset, logs, and analytics storage) • GPU: NVIDIA CUDA-enabled GPU
7.	**Methodology**	1) Data Collection — Collect quiz questions and educational datasets Store student performance data and analytics 2) Data Preprocessing — Clean and organize quiz datasets Categorize questions based on topics and difficulty levels 3) AI Model Development — Implement recommendation systems for personalized learning Develop adaptive learning algorithms for dynamic quiz generation Train predictive analytics models for performance analysis 4) Frontend Development — Develop responsive user interfaces using React.js and TypeScript Create dashboards for analytics and leaderboard visualization 5) Backend Development —Implement cloud database using Supabase PostgreSQL Develop APIs for quiz management, authentication, and analytics 6) Testing & Evaluation —Perform functionality testing and UI testing Evaluate AI model performance and system responsiveness
8.	**Expected Outcomes**	• Intelligent online assessment system with adaptive quizzes and AI-generated question support. • Real-time performance analytics dashboard for tracking student progress and learning behavior. • Secure online examination platform with anti-cheating and AI-based proctoring features. • Gamified learning system with leaderboards, badges, streaks, and multiplayer quiz competitions. • Cloud-based scalable educational platform suitable for schools, colleges, coaching centers, and training institutes.

9.	**Real-time Aplication**	• Online examination and assessment systems for schools and colleges • Real-time multiplayer quiz competition platforms • AI-based personalized learning and performance analytics systems • Secure remote assessment and online proctoring applications • E-learning and certification testing applications
10.	**Real-time Implementation Possibilities**	The proposed AI-Powered Adaptive Quiz and Performance Analytics Platform can be implemented in real-time in schools, colleges, coaching institutes, online learning platforms, and corporate training systems for conducting secure online assessments and personalized learning activities. The platform can be used for academic quizzes. Its AI-based adaptive learning, performance analytics, gamification, and anti-cheating features make it suitable for modern digital education environments. The cloud-based architecture and real-time dashboards also enable scalable deployment, remote accessibility, and efficient monitoring of student performance and learning progress.
11.	**Any other Information**	**Social Relevance:** • The project promotes accessible, intelligent, and personalized digital learning for students and educational institutions. **Economical Relevance:** • The platform reduces the cost and manual effort involved in traditional examination and evaluation systems. **Entrepreneurship Opportunity:** • The project has strong startup and SaaS potential for schools, colleges, coaching centers, and online learning platforms. **Innovative Idea:** • The system combines AI, adaptive learning, analytics, gamification, and secure assessments into a single intelligent platform.

		Learning Benefits: • The platform improves student engagement, performance tracking, and personalized learning experiences. **Technological Benefits:** • The project uses modern cloud and AI technologies to provide scalability, real-time analytics, and efficient performance monitoring.

Team Member Signatures **Guide Signature**

1. Hamsa R Gowda _____
2. Manasa N _____
3. Monika MU _____
4. Megha HR _____